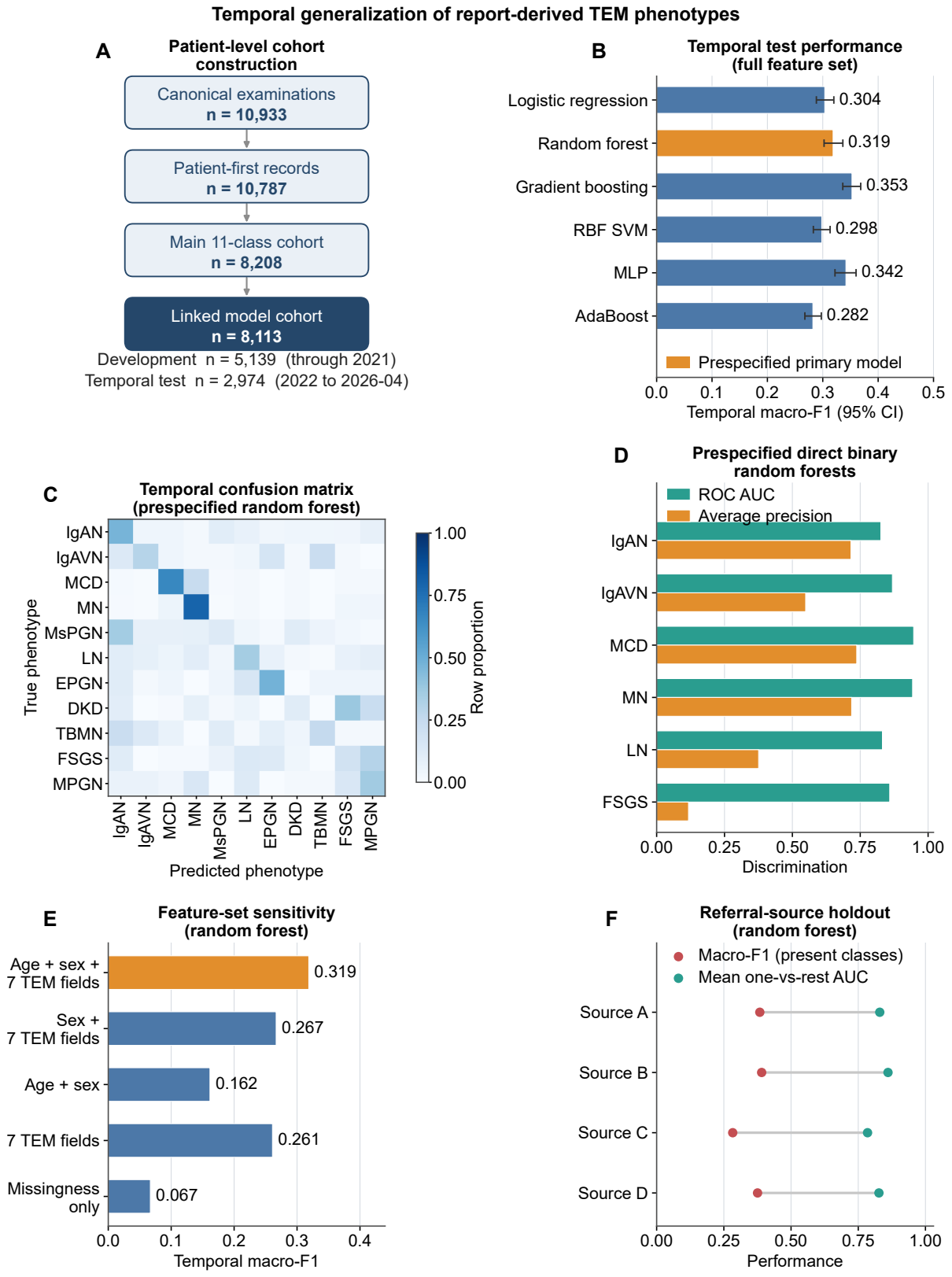

Temporal generalization of report-derived TEM phenotypes
A
Patient-level cohort construction
Canonical examinations
n = 10,933
Patient-first records
n = 10,787
Main 11-class cohort
n = 8,208
Linked model cohort
n = 8,113
Development n = 5,139 (through 2021)
Temporal test n = 2,974 (2022 to 2026-04)
B
Temporal test performance (full feature set)
Logistic regression
Random forest
Gradient boosting
RBF SVM
MLP
AdaBoost
0.304
0.319
0.353
0.298
0.342
0.282
Prespecified primary model
0.0
0.1
0.2
0.3
0.4
0.5
Temporal macro-F1 (95% CI)
C
Temporal confusion matrix (prespecified random forest)
IgAN
IgAVN
MCD
MN
MsPGN
LN
EPGN
DKD
TBMN
FSGS
MPGN
True phenotype
Predicted phenotype
1.00
0.75
0.50
0.25
0.00
Row proportion
D
Prespecified direct binary random forests
ROC AUC
Average precision
IgAN
IgAVN
MCD
MN
LN
FSGS
0.00
0.25
0.50
0.75
1.00
Discrimination
E
Feature-set sensitivity (random forest)
Age + sex + 7 TEM fields
Sex + 7 TEM fields
Age + sex
7 TEM fields
Missingness only
0.319
0.267
0.162
0.261
0.067
0.0
0.1
0.2
0.3
0.4
Temporal macro-F1
F
Referral-source holdout (random forest)
Macro-F1 (present classes)
Mean one-vs-rest AUC
Source A
Source B
Source C
Source D
0.00
0.25
0.50
0.75
1.00
Performance